Character LCD Module Controller (VHDL) - eewiki / Logic - Engineering and Component Solution Forum - TechForum │ Digi-Key

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Code Download

Version 3.0: lcd_controller.vhd (6.4 KB)
Added LCD configuration using generics

Version 2.0: lcd_controller_v2.vhd (8.0 KB)

Version 1.0 is no longer available.

Example that instantiates the lcd_controller.vhd component and uses it to write “123456789” to an lcd module: lcd_example.vhd (3.3 KB)

Introduction

This LCD controller is a VHDL component for use in CPLDs and FPGAs. The controller manages the initialization and data flow to HD44780 compatible 8-bit interface character LCD modules. It was primarily developed pursuant to the Lumex LCD General Information datasheet. This example VHDL component allows simple LCD integration into practically any programmable logic application. Figure 1 depicts the controller implemented to interface between an LCD module and a user’s custom logic.

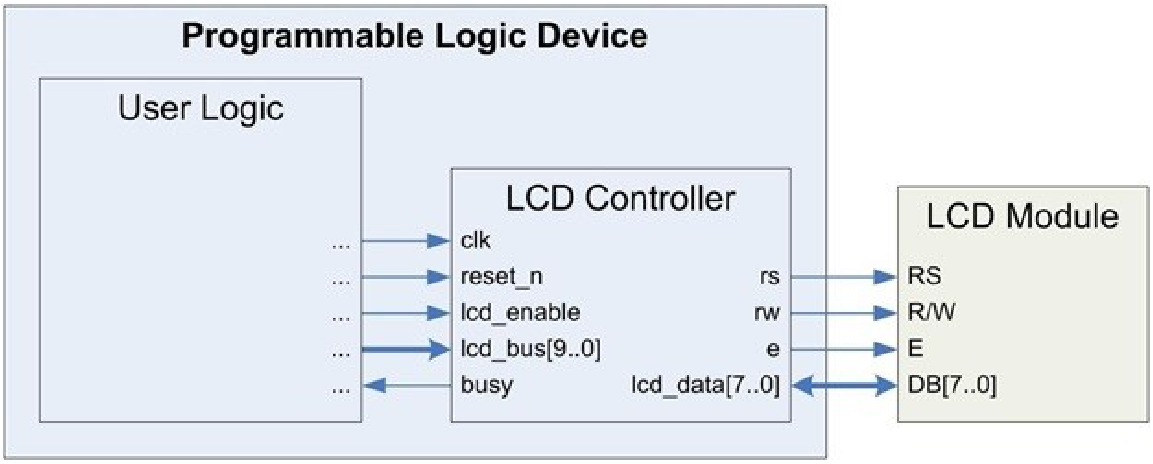

Figure 1. LCD Controller Implementation in PLD

State Machine

The LCD controller state machine consists of five states. Upon startup, it immediately enters the Power-up state, where it waits 50ms to ensure the supply voltage has stabilized. It then proceeds to an Initialize state. The controller cycles the LCD through its initialization sequence, setting the LCD’s parameters to default values defined in the hardware. This process completes in approximately 2.2ms, and the controller subsequently assumes a Ready state. It waits in this state until the lcd_enable input is asserted, then advances to the Send state. Here, it communicates the appropriate information to the LCD, as defined by the lcd_bus input. After 50us, it returns to the Ready state until further notice. If a low logic level is applied to the reset_n input at any time for a minimum of one clock cycle, the controller resets to the Power-up state and re-initializes. Figure 2 illustrates the LCD controller state machine.

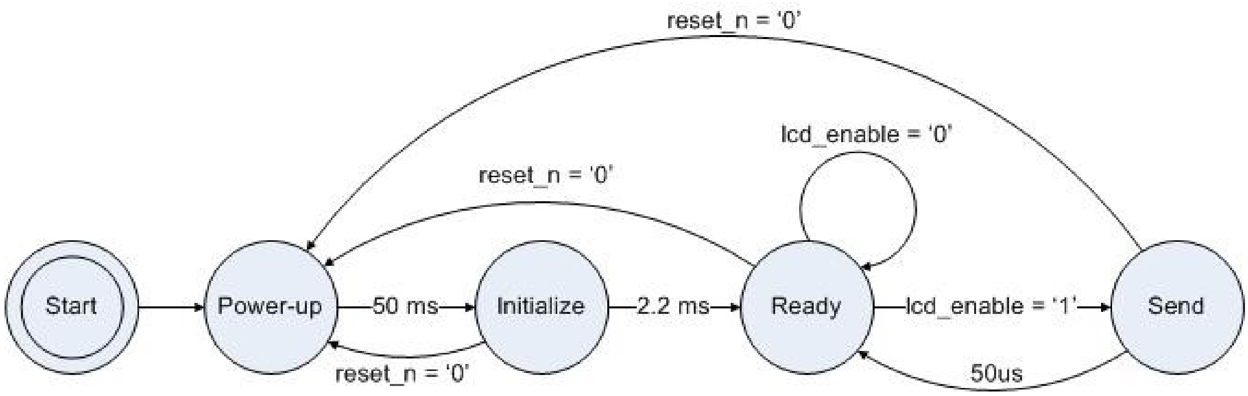

Figure 2. LCD Controller State Machine

Port Descriptions

Table 1 describes the LCD controller's interface.

Table 1. LCD Controller I/O Description

I/O Name	Width	Mode	Description	Interface
clk	1	input	Clock for LCD controller. Default set for 50MHz. If a different frequency is desired, change the constant freq in the architecture declarations to reflect the new frequency in MHz.	system clock
reset_n	1	input	Active low synchronous reset pin. This pin must be set high to implement the LCD controller. Setting the pin low for one or more clock cycles restarts the LCD controller state machine.	user logic
lcd_enable	1	input	Data latch for LCD controller. H: initiates a transaction using the data currently on the lcd_bus, L: no transaction is initiated and any data on lcd_bus is ignored	user logic
lcd_bus	10	input	Data/instructions to be sent to the LCD module. The MSB is the rs signal, followed by the rw signal. The other 8 bits are the data bits. The LSB on the bus corresponds to the least significant data bit.	user logic
busy	1	output	Feedback on the state of the LCD controller. H: the controller is busy initializing or conducting a transaction with the LCD module, any instructions/data sent will be ignored, L: the controller is idle and ready to accept commands for a transaction	user logic
rs	1	output	LCD module Register Select Signal; H: sending data, L: sending instructions	LCD pin 4
rw	1	output	LCD module Read/Write Select Signal; H: Read, L: Write	LCD pin 5
e	1	output	LCD module enable signal	LCD pin 6
lcd_data	8	bidir	Data bus to the LCD module / busy signal from the LCD	LCD pins 7-14

Initialization

The LCD controller executes an initialization sequence each time it is powered-up or the reset_n pin is deasserted for a minimum of one clock cycle. The controller asserts the busy pin during initialization. Once initialization completes, the busy pin deasserts, and the LCD controller waits in the Ready state for input from the user logic.

The initialization sequence specifies several LCD parameters through a series of four commands:

- Function Set (sets 4-bit/8-bit display, number of display lines, and character font)
- Display On/Off Control (turns on/off the display, the cursor, and cursor blinking)
- Display Clear (clears the display)
- Entry Mode Set (sets increment/decrement mode, and turns on/off shifting)

The parameters values are specified by setting the GENENIC parameters in the ENTITY. Table 2 describes the parameters. The user can also send commands to the LCD to change any parameters after initialization.

Table 2. Generic Parameters Descriptions

Generic	Data Type	Default	Description
clk_freq	integer	50	Frequency of the system clock input (PORT clk) in MHz
display_lines	standard logic	1	Number of display lines: 0 = 1-line mode, 1 = 2-line mode
character_font	standard logic	0	Font: 0 = 5x8 dots, 1 = 5x10 dots
display_on_off	standard logic	1	Display on/off: 0 = off, 1 = on
cursor	standard logic	0	Cursor on/off: 0 = off, 1 = on
blink	standard logic	0	Blink on/off: 0 = off, 1 = on
inc_dec	standard logic	1	Increment/decrement: 0 = decrement, 1 = increment
shift	standard logic	0	Shift on/off: 0 = off, 1 = on

Transactions

Upon deassertion of the busy pin, the LCD controller enters the Ready state. The user logic can interface via the lcd_enable and lcd_bus pins to conduct transactions with the LCD module. The user initiates this process by issuing the desired data/instruction to the lcd_bus and asserting the lcd_enable pin. The LCD controller then asserts the busy pin and manages the transaction. When finished, the controller deasserts the busy pin, indicating that it is ready for another instruction. Figure 3 depicts the timing diagram for the beginning of a transaction.

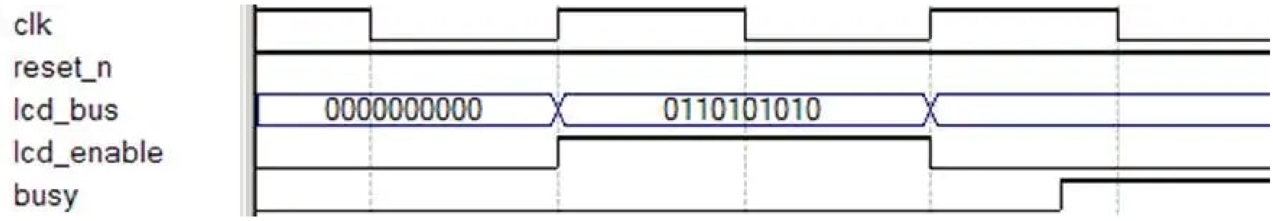

Figure 3. Transaction Timing Diagram

Conclusion

The LCD control logic provided manages the initialization and data flow between custom user logic and the 8-bit interface mode of HD44780 compatible character LCD modules. The user can set the system clock frequency and change the default initialization parameters by setting GENERIC parameters in the ENTITY.

Additional Information